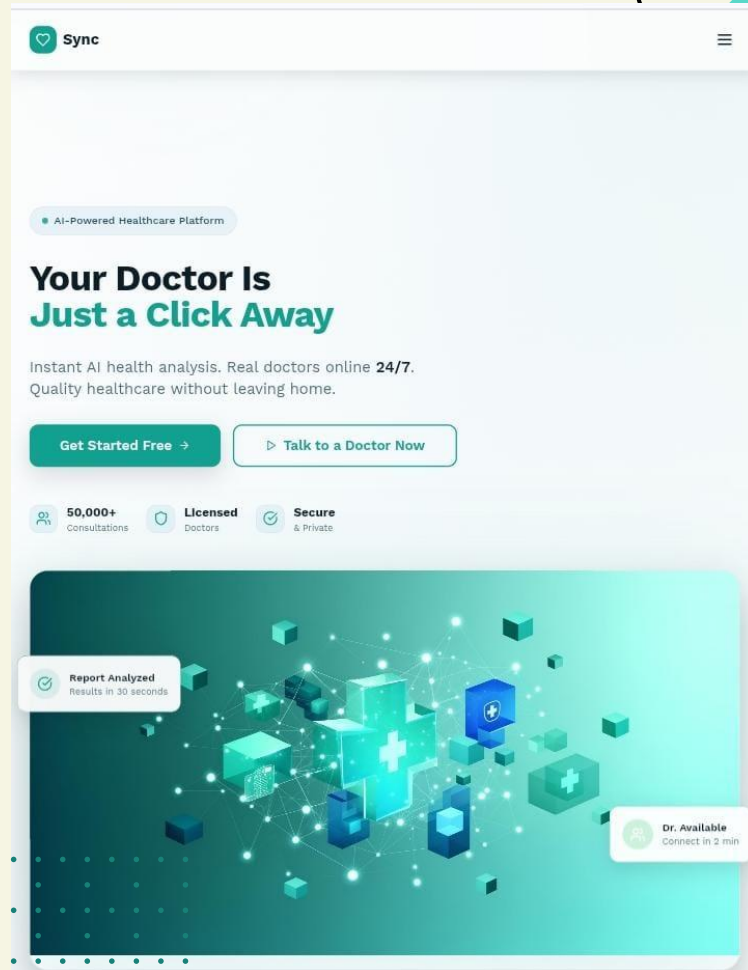


+ BHG-AI-(25-26)

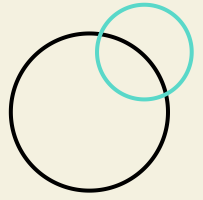


BRIDGING HEALTHCARE GAPS WITH AI

FINAL YEAR PROJECT

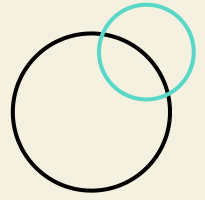


CONTENTS OF THIS PRESENTATION



1. **OBJECTIVES OF THE PROJECT**
2. **EXISTING PLATFORMS VS HEALTHBRIDGE.AI**
3. **CORE FEATURES OF OUR PROJECT**
4. **AI HEALTH ASSISTANT**
5. **MEDICAL REPORT ANALYSIS**
6. **TELECONSULTATION SYSTEM**
7. **DOCTOR DIRECTORY AND MATCHING**
8. **TECHNOLOGY STACK**
9. **IMPLEMENTATION CHALLENGES AND SOLUTIONS**
10. **SECURITY AND PRIVACY MEASURES**
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GROUP MEMBERS

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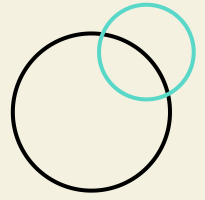
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Project Manager:

Prof. Dr. Muhammad Ilyas



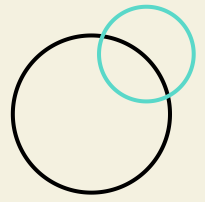


PROBLEM STATEMENT

The Healthcare Crisis in Pakistan:

- Limited access to medical professionals, especially in rural areas.
- Long waiting times at hospitals.
- High consultation costs for underserved populations.
- Language barriers and low health literacy.

Healthcare challenge	Statistics	Source/Reference
Critical Doctor Shortage	1:1000 doctor to patient ratio	PMC, 2023
High out of pocket costs	60% of health expenses directly paid	JPMA, 2025
Rural access gap	70% population VS 30% healthcare facilities	World Bank, 2024



PROBLEM STATEMENT

Healthcare challenge	Statistics	Source/Reference
Low health literacy	Only 38% understand medical instructions	National Health Survey, 2023
Telemedicine lag	<15% providers offer online consultation	Pakistani telehealth, 2024

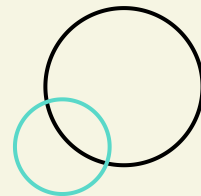
Result: 45% of Pakistanis skip medical care due to costs and distance.

SOLUTION

An AI-powered platform making healthcare accessible, affordable, and understandable.



EXISTING PLATFORMS VS HEALTHBRIDGE.AI



Existing Platforms

1.

Oladoc

Online doctor
appointment
booking platform.

2.

Marham

Telemedicine and
online consultations.

3.

Sehat Kahani

Video-based digital
healthcare services.

HealthBridge.ai

HealthBridge.ai uses AI to explain medical reports, symptoms, and terms in simple Urdu/English, helping patients understand their health independently before visiting a doctor.

OBJECTIVES OF THE PROJECT

1.

AI Symptom Checker

2.

AI Report Scanner

3.

Video Consultation Platform

4.

Doctor-patient Matching

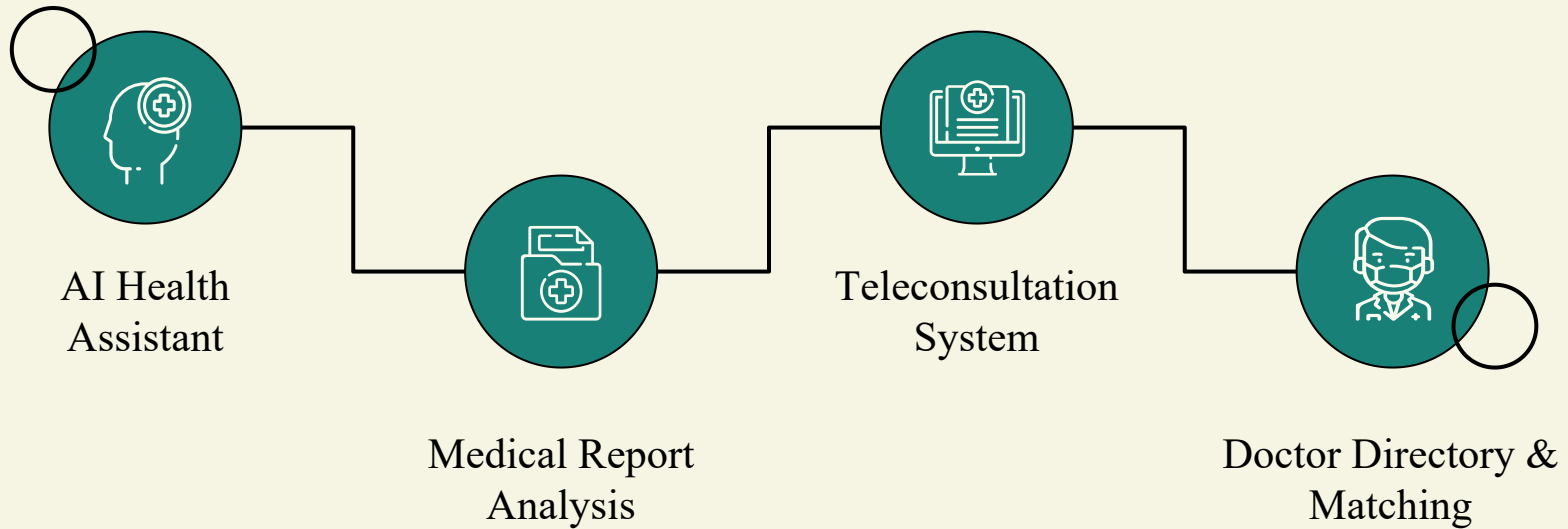
5.

**Medicine Tracker &
Reminders**

6.

Simple & Accessible Design

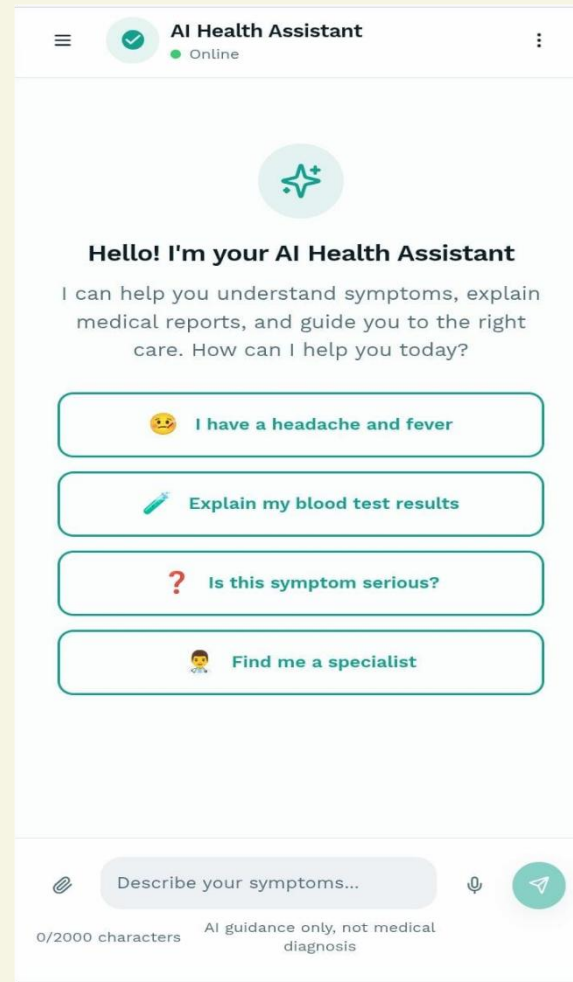
CORE FEATURES





1. AI Health Assistant

- Symptom analysis and triage (4 urgency levels)
- Bilingual support (English/Urdu)
- Context-aware conversations





2. Medical Report Analysis

- Upload reports (PDF/JPG/PNG)
- OCR text extraction
- Plain language explanations



3. Teleconsultation System

- Video/audio consultations via WebRTC
- Real-time chat during sessions
- Appointment booking with reminders

The screenshot shows a mobile application interface for booking an appointment. At the top, there is a back arrow and the title 'Book Appointment'. Below this is a progress indicator for 'Step 2 of 3: Consultation Details'. The main content area includes a date and time selection box showing 'January 7, 2025 at 09:30 AM' and the doctor's name 'Dr. Bilal Hussain • General Physician'. Under the heading 'Consultation Type', there are four buttons: 'Video Call', 'Audio Call', 'Text Chat', and 'Physically Visit'. The 'Physically Visit' button is highlighted with a green border and a checkmark icon. Below this is a 'Reason for Visit' section with a text input field containing the word 'test' and a character count '4/200'. There is also a checkbox option 'Share my health records with doctor before consultation'. At the bottom, there is an 'Attach Reports' section with a note '(Optional)' and two buttons: 'Back' and 'Continue →'.

Book Appointment

Step 2 of 3: Consultation Details

January 7, 2025 at 09:30 AM
Dr. Bilal Hussain • General Physician

Consultation Type

Video Call Audio Call Text Chat Physically Visit

Reason for Visit *

test 4/200

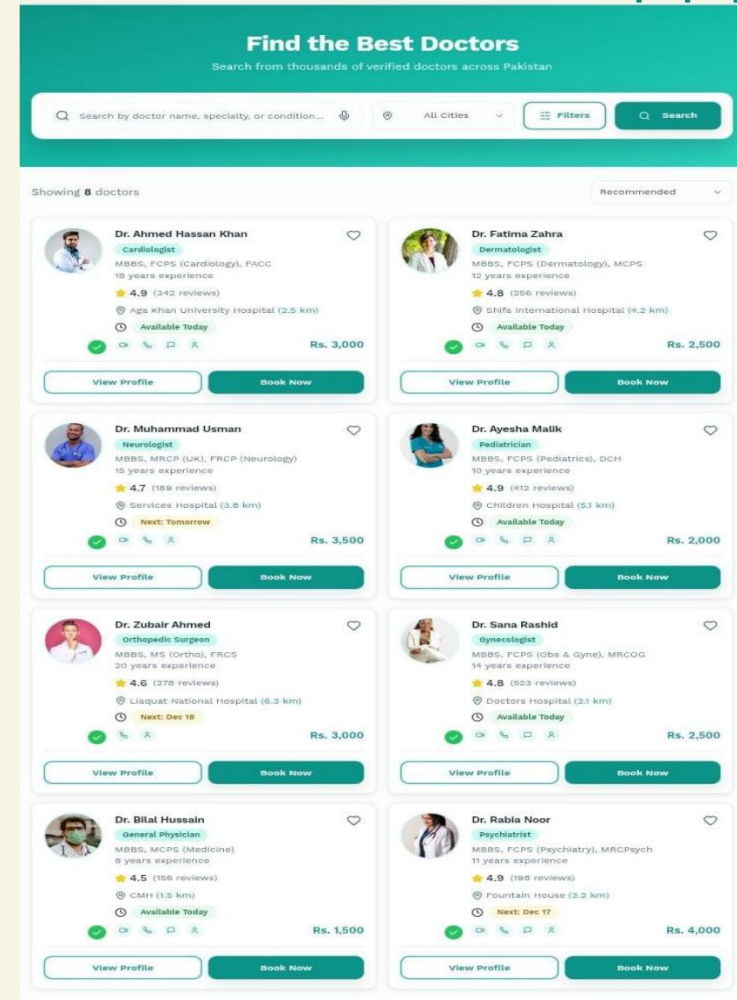
☐ Share my health records with doctor before consultation

Attach Reports (Optional)

Back Continue →

4. Doctor Directory & Matching

- Search by specialty, location, availability
- Verified doctor profiles
- Payment integration (JazzCash, EasyPaisha, Bank transfers)





5. Doctor Profile Page

- Complete doctor information including name, photo, qualifications
- Rating system with patient reviews
- Verification from Pakistan Medical Commission
- Multiple consultation modes: Video, Audio, Chat, and In-Person
- Educational background and professional certifications
- Areas of expertise
- Hospital affiliation
- Clinic details with full address, contact number, and weekly schedule
- Available facilities
- Two action buttons: "Message Dr." and "Book Appointment" for patient engagement

The screenshot displays a mobile app interface for a doctor's profile. At the top, the name 'Dr. Bilal Hussain' is shown with a back arrow, heart, and message icon. Below this is a profile card featuring a circular photo of the doctor, a green verification checkmark, and text: 'Dr. Bilal Hussain', 'MBBS, MCPs (Medicine) • General Physician', '4.5 (156 reviews)', 'Family Care Clinic, Rawalpindi', '8 years experience', 'Verified by Pakistan Medical Commission', and consultation options: 'Video', 'Audio', 'Chat', and 'In-Person'.

The 'About' section contains a paragraph: 'Dr. Bilal Hussain is a dedicated general physician with special interest in managing chronic diseases like diabetes and hypertension.'

The 'Education & Qualifications' section lists: 'MBBS, Army Medical College, 2012' and 'MCPs (Medicine), College of Physicians & Surgeons Pakistan, 2018'.

The 'Areas of Expertise' section shows tags for 'Diabetes', 'Hypertension', 'General Check-ups', and 'Fever & Infections'.

The 'Hospital Affiliations' section lists 'Combined Military Hospital Rawalpindi'.

The 'Certifications' section lists 'Diabetes Management Course' and 'Critical Care Training'.

The 'Clinic Details' section includes: 'Family Care Clinic, Saddar, CMH Complex, Rawalpindi', '+92-31-92708001', and a table of working hours.

Day	Working Hours
Monday	9:00 AM - 9:00 PM
Tuesday	9:00 AM - 9:00 PM
Wednesday	9:00 AM - 9:00 PM
Thursday	9:00 AM - 9:00 PM
Friday	9:00 AM - 9:00 PM
Saturday	9:00 AM - 9:00 PM
Sunday	10:00 AM - 6:00 PM

The 'Facilities' section lists 'Parking', 'Lab Services', and 'Pharmacy'.

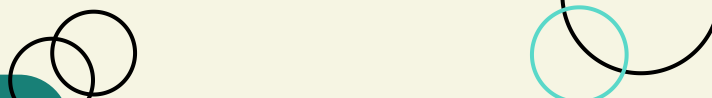
The 'Patient Reviews' section shows a 4.5 star rating from 156 reviews, with a bar chart showing the distribution of ratings. A sample review from 'Hassan Ali' dated '2024-01-11' with a 5-star rating is shown.

At the bottom, there are two buttons: 'Message Dr.' and 'Book Appointment', with a 'Log in to Book' link.



6. Book Appointment Page

- Doctor's name displayed in header with back navigation, favorites, and chat icons
- Consultation fee clearly shown
- Weekly availability calendar showing dates
- Available dates highlighted in teal, unavailable dates in light pink/gray
- Two prominent action buttons: "Book Appointment" (primary) and "Message Doctor" (secondary)
- Cancellation policy: Free cancellation up to 1 hour before appointment
- Accepted payment methods: Cash and JazzCash
- Languages spoken by doctor: English, Urdu, and Punjabi
- Fixed bottom navigation with "Message Dr." and "Book Appointment" buttons for easy access



Technology Stack

Frontend

- React.js, Next.js, TypeScript
- WebSocket for real-time updates
- PWA for mobile experience

Backend

- Node.js, Express.js
- JWT Authentication, REST APIs
- WebRTC for video calls

AI/ML

- Python, TensorFlow/PyTorch
- BERT for NLP, Google Vision OCR
- Medical knowledge graphs

Database

- PostgreSQL, Redis
- Google Cloud Platform
- Docker containers

IMPLEMENTATION CHALLENGES & SOLUTIONS



Challenge 1

AI Accuracy in Medical Context



Solution

- Confidence thresholding
- Human escalation
- Multi-model ensemble

IMPLEMENTATION CHALLENGES & SOLUTIONS



Challenge 2

Network Issues in Rural Pakistan



Solution

- Adaptive bitrate streaming
- Audio-only fallback
- Offline capabilities

IMPLEMENTATION CHALLENGES & SOLUTIONS



Challenge 3

OCR Accuracy on Medical Reports



Solution

- Image preprocessing
- Manual correction interface
- Confidence scoring

IMPLEMENTATION CHALLENGES & SOLUTIONS



Challenge 4

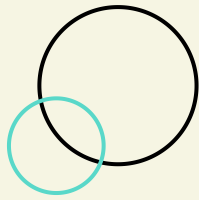
Payment Gateway Reliability



Solution

- Adapter pattern
- Retry mechanisms
- Multiple provider support

SECURITY & PRIVACY MEASURES



Data Protection

- End-to-end encryption for sensitive data
- JWT authentication with refresh tokens
- Role-Based Access Control (RBAC)



Compliance

- Pakistan Medical Commission guidelines
- Data privacy regulations compliance
- Medical disclaimers and user consent



Security Features

- HTTPS-only communication
- Input validation and sanitization
- Comprehensive audit logging

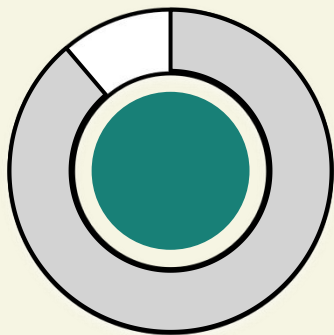
RESULTS & IMPACT : Expected Outcomes

- 1 Reduced healthcare access barriers in rural areas
- 2 Lower consultation costs for underserved populations
- 3 Improved health literacy through AI explanations
- 4 Time savings for both patients and doctors



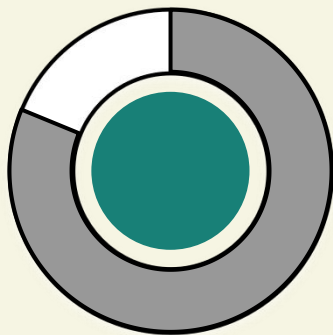
TECHNICAL METRICS

95%



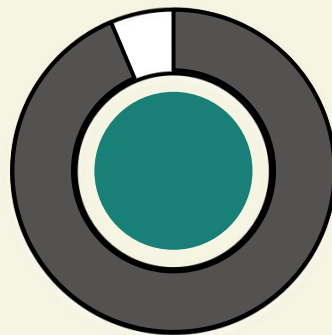
AI response time: < 5 seconds (95% faster than traditional triage)

90%



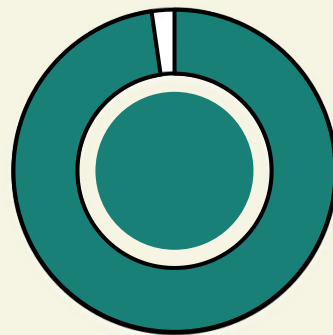
OCR accuracy: > 90% on standard reports

95%



System availability: > 95%

99%



Concurrent users supported: 1000+ (99% request success rate under peak load of 1,000+ concurrent users)

FUTURE ENHANCEMENTS

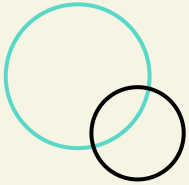


SHORT-TERM

- Integration with wearable devices
- Emergency services coordination
- Prescription management

LONG-TERM

- Chronic disease prediction models
- Integration with hospital EHR systems
- AI-powered diagnostic imaging analysis
- Mental health support modules



CONCLUSIONS



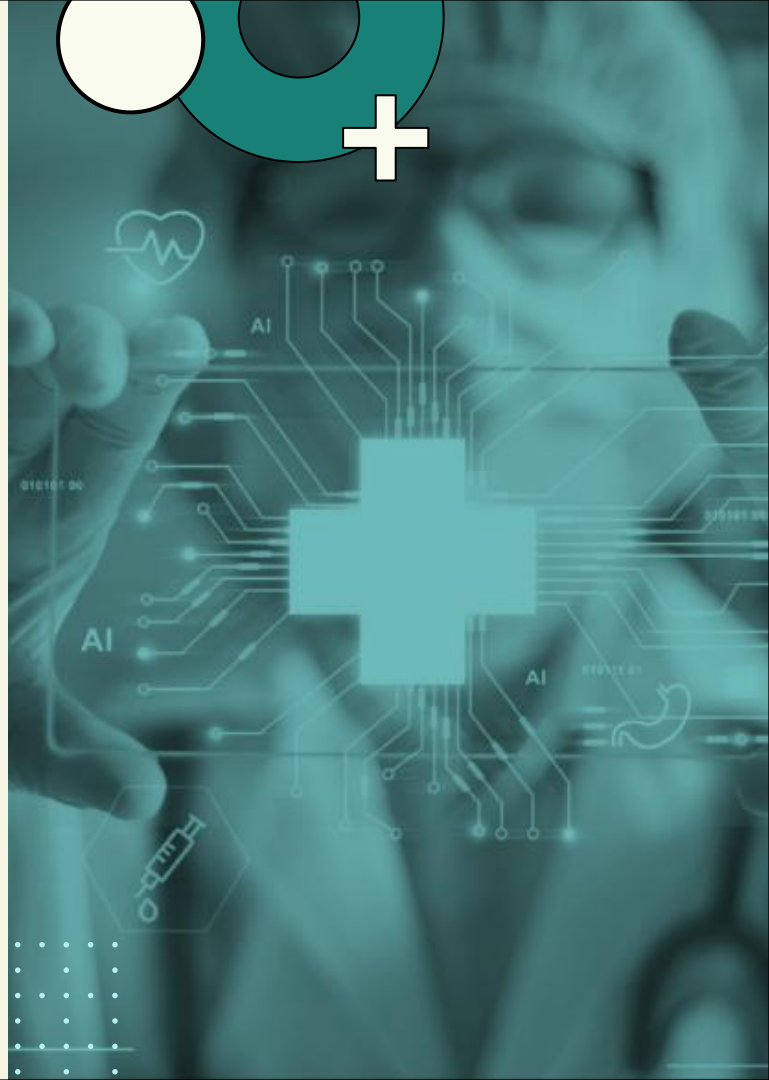
For Patients

Provides accessible, affordable, and understandable healthcare guidance and consultations anytime, anywhere



For Doctors

Enables expanded reach and more efficient consultations, optimizing their time and practice.



CONCLUSIONS

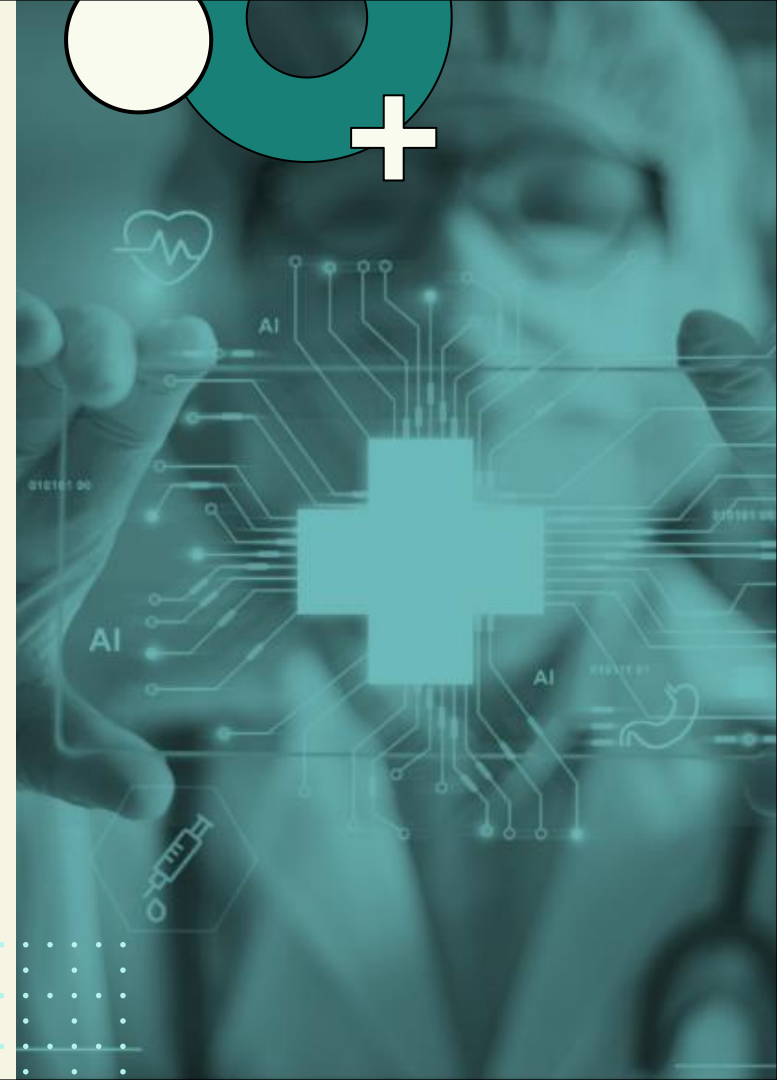


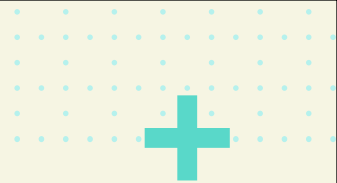
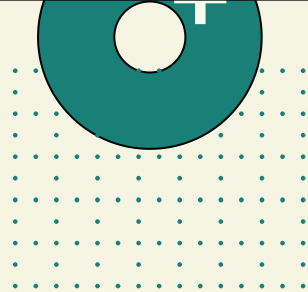
For Society

Contributes to reduced healthcare disparities by bridging gaps in access and affordability.



This integrated platform represents a significant step toward democratizing quality healthcare in Pakistan, leveraging technology to create a more equitable and efficient ecosystem for all stakeholders.





THANK YOU !

